

Basic Principles of Medicine 2

Module: Digestion and Metabolism

DM 10: Glycogen Metabolism and Glycogen Storage Diseases

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Purpose of pathway: Liver and muscle
Regulation: Fed vs fasting; Liver and muscle
Disorders

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Objectives: Glycogen Metabolism and Glycogen Storage Diseases

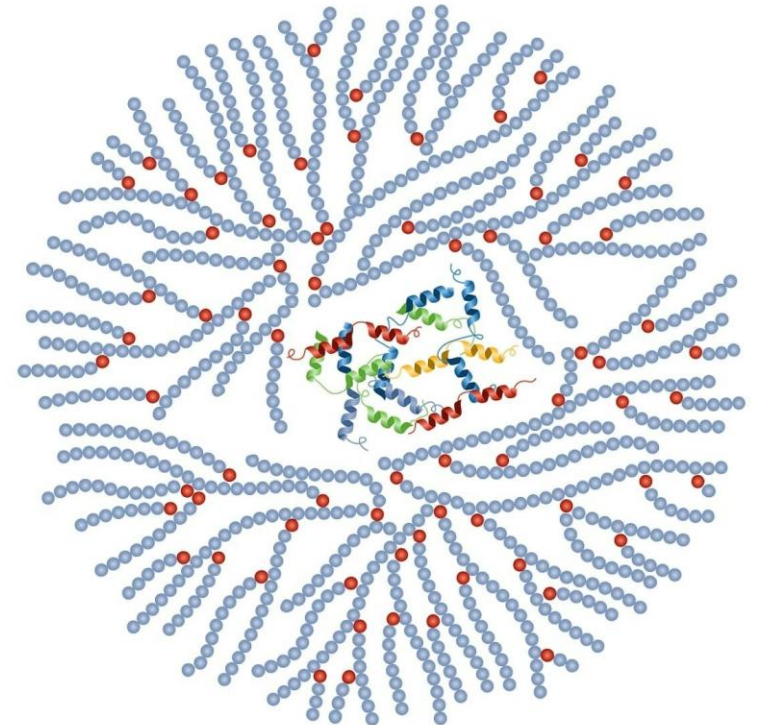
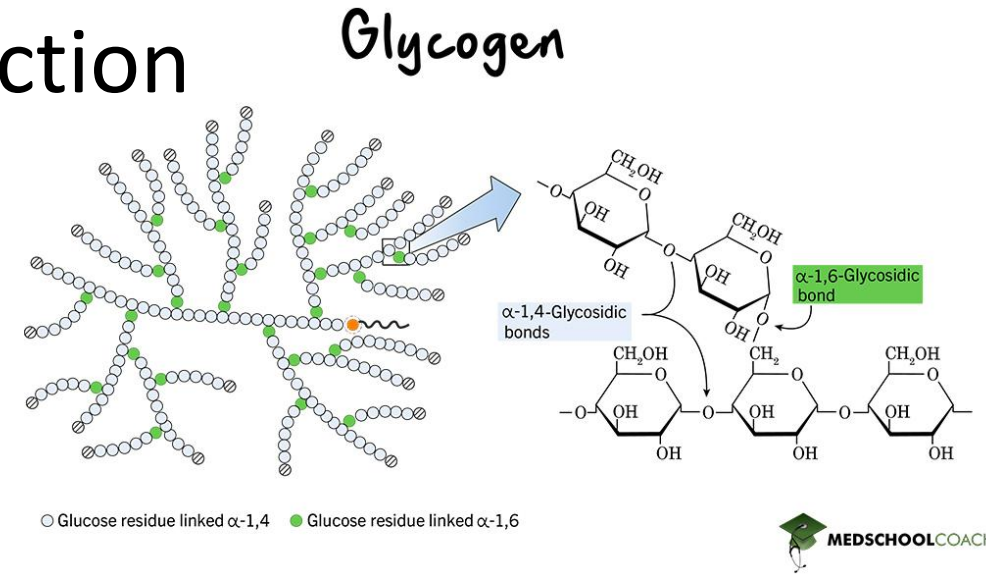
SOM. MK.I.BPM2.3.DM.2.BCHM.1154	Describe the structure-function relationship of glycogen in tissues that store glycogen.
SOM. MK.I.BPM2.3.DM.2.BCHM.1155	Compare and contrast the purpose of glycogen metabolism in skeletal muscle and liver.
SOM. MK.I.BPM2.3.DM.2.BCHM.1156	Describe the role of insulin in regulation of glycogenesis (glycogen synthase) in liver and muscle.
SOM. MK.I.BPM2.3.DM.2.BCHM.1157	Outline glycogenesis and review the significance of glycogen synthase and the branching enzyme.
SOM. MK.I.BPM2.3.DM.2.BCHM.1158	Compare and contrast glycogen degradation in liver and muscle with reference to: hormones, metabolic state, and fate of glucose 6-P.
SOM. MK.I.BPM2.3.DM.2.BCHM.1159	Outline glycogenolysis and highlight importance of glycogen phosphorylase and debranching enzyme.
SOM. MK.I.BPM2.3.DM.2.BCHM.1160	Evaluate the role of insulin and glucagon in reciprocal regulation of carbohydrate metabolism in the liver.
SOM. MK.I.BPM2.3.DM.2.BCHM.1161	Compare and contrast carbohydrate metabolism in the liver during fed and fasting state.
SOM. MK.I.BPM2.3.DM.2.BCHM.1162	List the glycogen storage diseases (Types I through VII) and be able to differentiate the disorders based on clinical and laboratory findings.
SOM. MK.I.BPM2.3.DM.2.BCHM.1163	Describe type I (von Gierke) glycogen storage disease including the metabolic basis for severe hypoglycemia, hepatomegaly and lactic acidosis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1164	Describe type II (Pompe disease) and outline lysosomal glycogen degradation
SOM. MK.I.BPM2.3.DM.2.BCHM.1165	Differentiate between glycogen storage diseases type III (Cori) and IV (Andersen) based on the abnormal structure of glycogen.
SOM. MK.I.BPM2.3.DM.2.BCHM.1166	Explain basis for clinical manifestations of type V (McArdle) glycogen storage disease: decreased exercise tolerance and lack of lactate increase in blood after exercise.

Recommended Reading

- Lippincott's Biochemistry 9th Edition; Chapter 11: Glycogen metabolism
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- Lippincott Illustrated Reviews (**Lippincott**) 9th edition: Chapter 11: Pages 137–148
[Glycogen Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Study questions in Chapter 11: 11.1 -11.7
- Videos:
 - [Glycogen metabolism - YouTube](#)
 - [Metabolism | Glycogenesis – YouTube](#)
 - [Metabolism | Glycogenolysis – YouTube](#)
 - [Metabolism | Regulation of Glycogen Metabolism - YouTube](#)

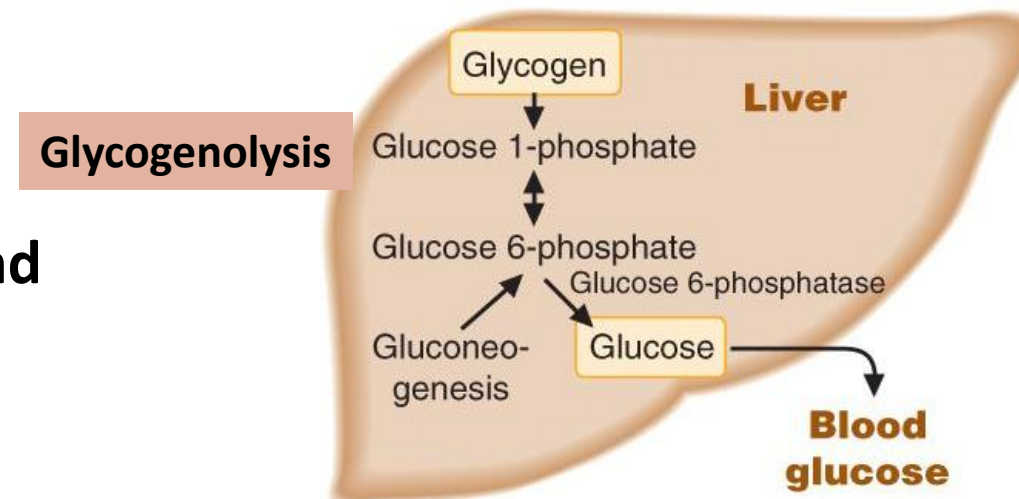
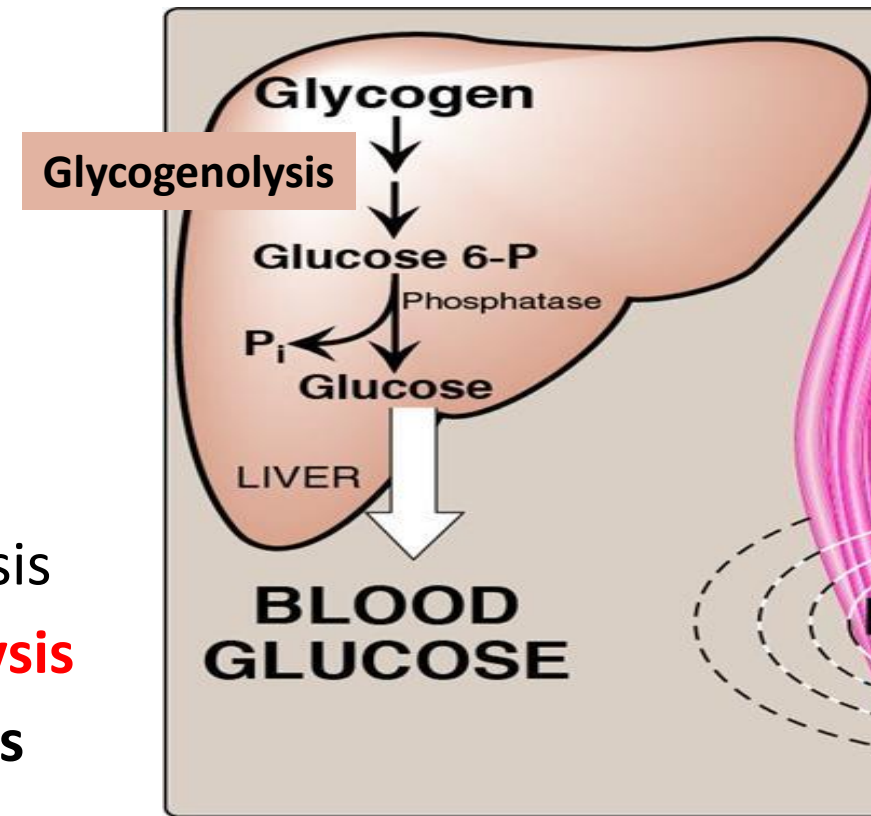
Glycogen: Structure, location and function

- Highly branched storage homopolysaccharide (Glucose)
 - Compare with starch and cellulose
- Cytosol (glycogen granules)
- Bonds: Enzymes are specific for the bonds
 - Alpha1→4 bonds in main chain;
 - Alpha 1→6 bonds at branch points
- LIVER and MUSCLE (skeletal and cardiac)
- **Function is different in liver and muscle**
- **Regulation is different in liver and muscle**
- **Differential diagnosis of Glycogen storage disorders**



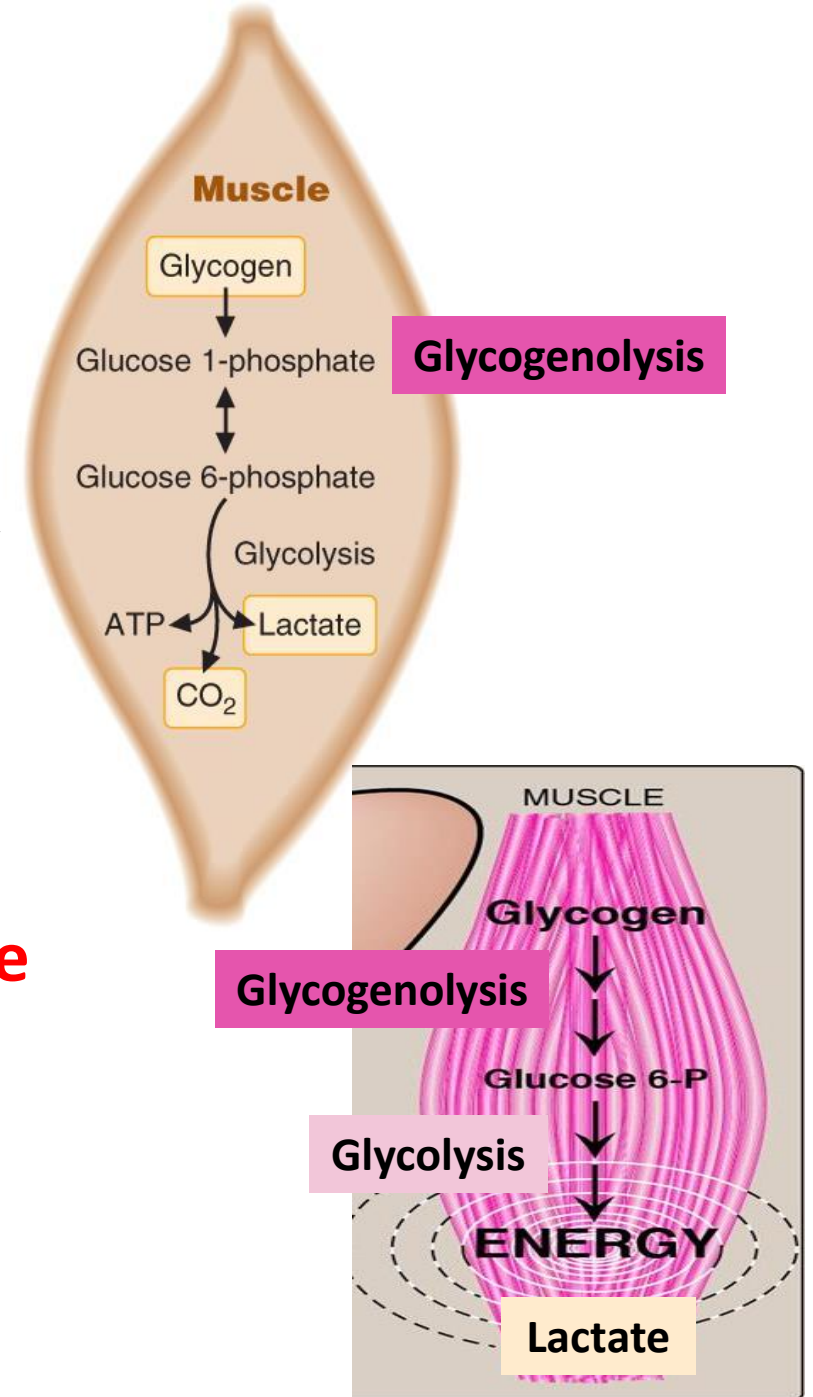
Liver Glycogen

- Maintains **plasma glucose level**
 - Glycogen → Glucose 6-P → Glucose
- Regulation by insulin: glucagon ratio
- High insulin: glucagon ratio (fed state) facilitates glycogenesis
- **Low insulin: glucagon ratio (fasting) facilitates glycogenolysis**
- **Blood glucose levels increase following liver glycogenolysis**
- Epinephrine also activates glycogenolysis
- Liver glycogen content fluctuates
 - Increases in well-fed state
 - Depleted in fasting
- **Liver glycogen storage disorders → Hypoglycemia and Hepatomegaly**



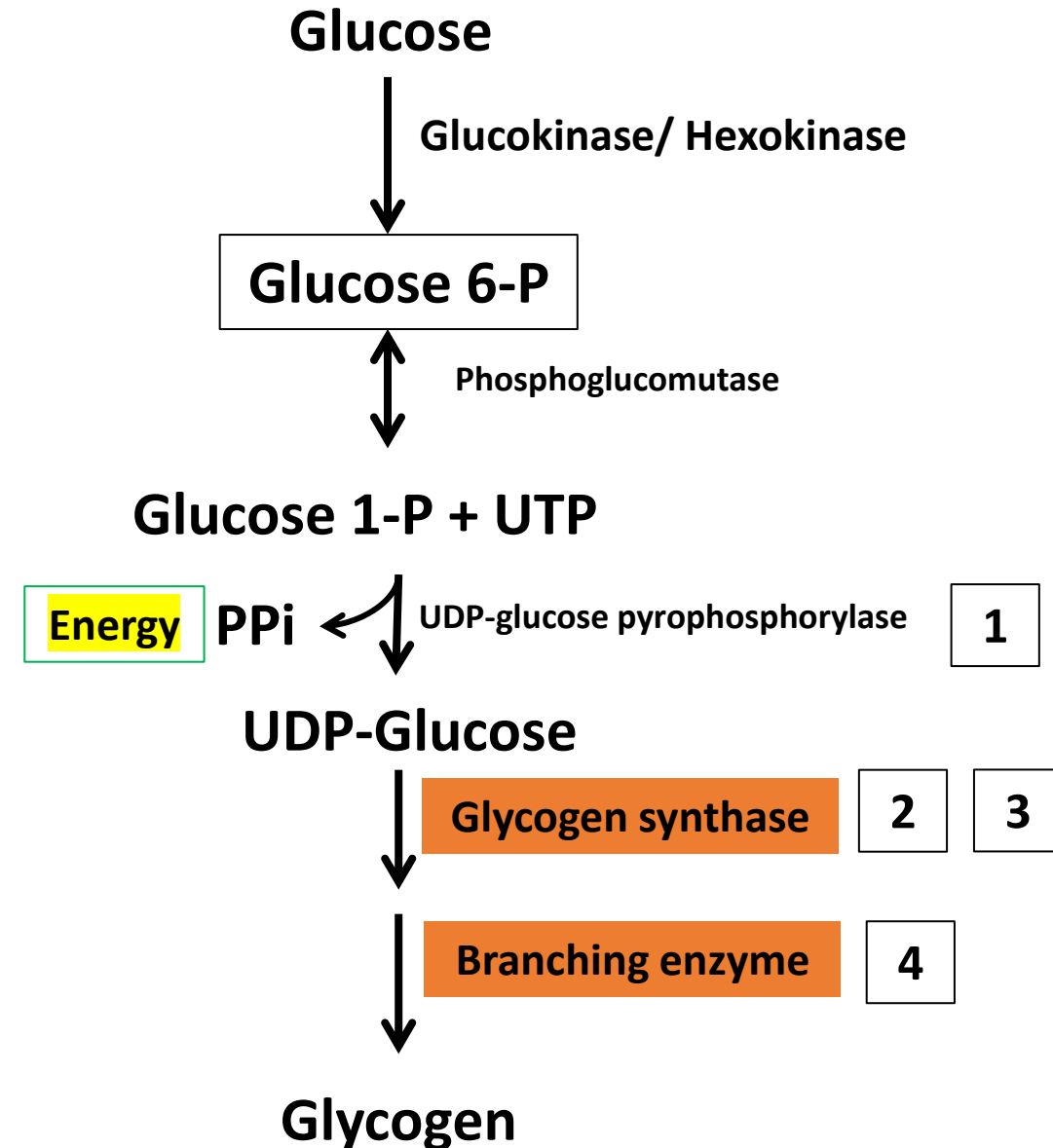
Muscle Glycogen

- Provides energy for **muscle contraction**
 - Anaerobic/ Strenuous: Glycogen → Glucose 6-P → Pyruvate → **Lactate**
 - Aerobic and Cardiac muscle: Glycogen → Glucose 6-P → Pyruvate → Acetyl CoA → Krebs cycle and ETC
- **Blood lactate levels increase following strenuous exercise**
- High insulin: glucagon ratio facilitates glycogenesis
- **Exercise (AMP and Ca²⁺) and epinephrine facilitate glycogenolysis**
- **Muscle glycogen storage disorders → Exercise intolerance, fatigue, cramps and muscle pain**
- Muscle glycogenolysis does NOT increase blood glucose level



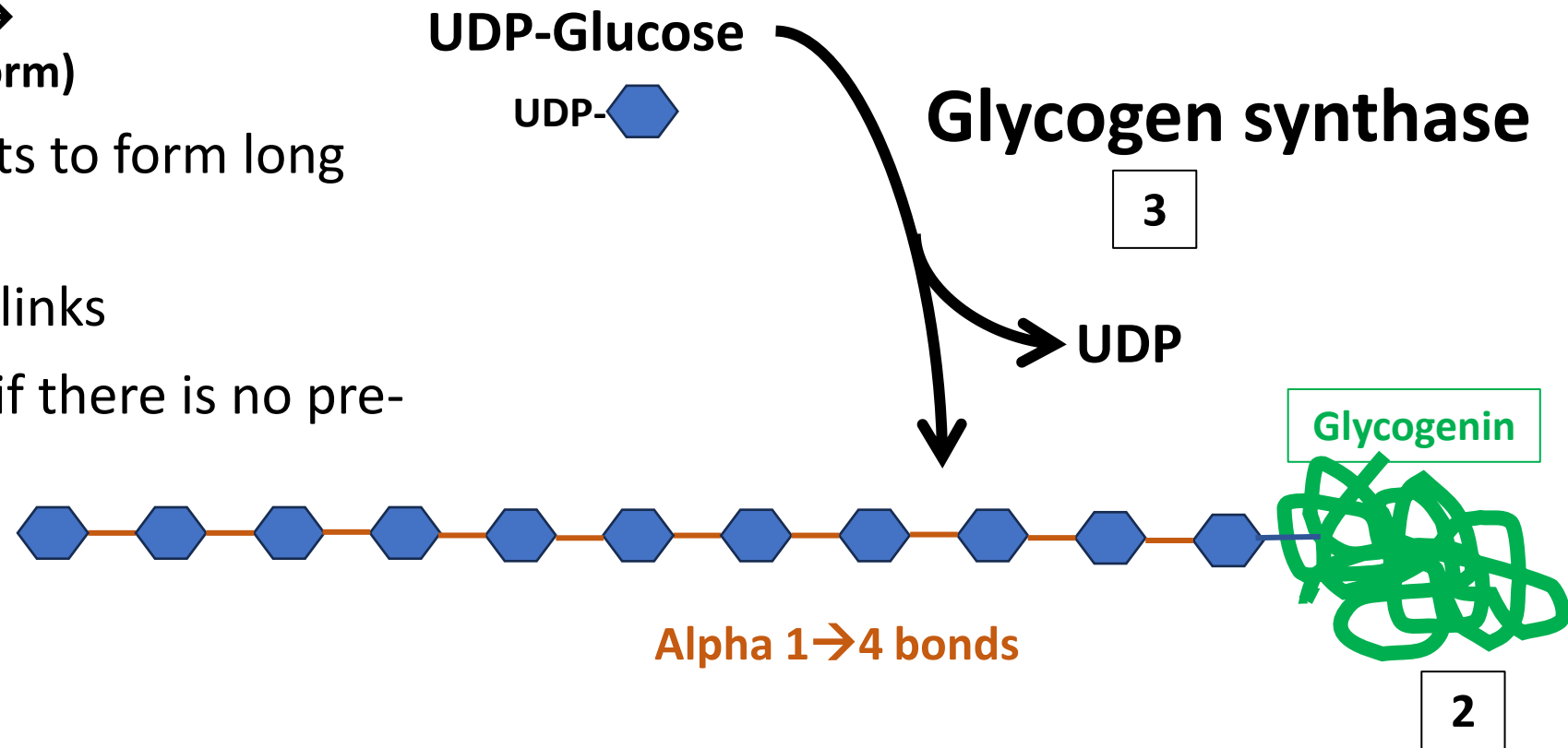
Glycogenesis (Glycogen synthesis)

- Process similar in liver and muscle
- Activation of glucose units
 - Formation of UDP-glucose 1
- Glycogenin (primer) 2
- Formation of long chains of glucose
 - **Glycogen synthase** 3
 - Regulated enzyme
 - Creates 1→4 bonds
- Formation of branches
 - Branching enzyme 4
 - Creates alpha 1→6 bonds
- [Metabolism | Glycogenesis - YouTube](#)

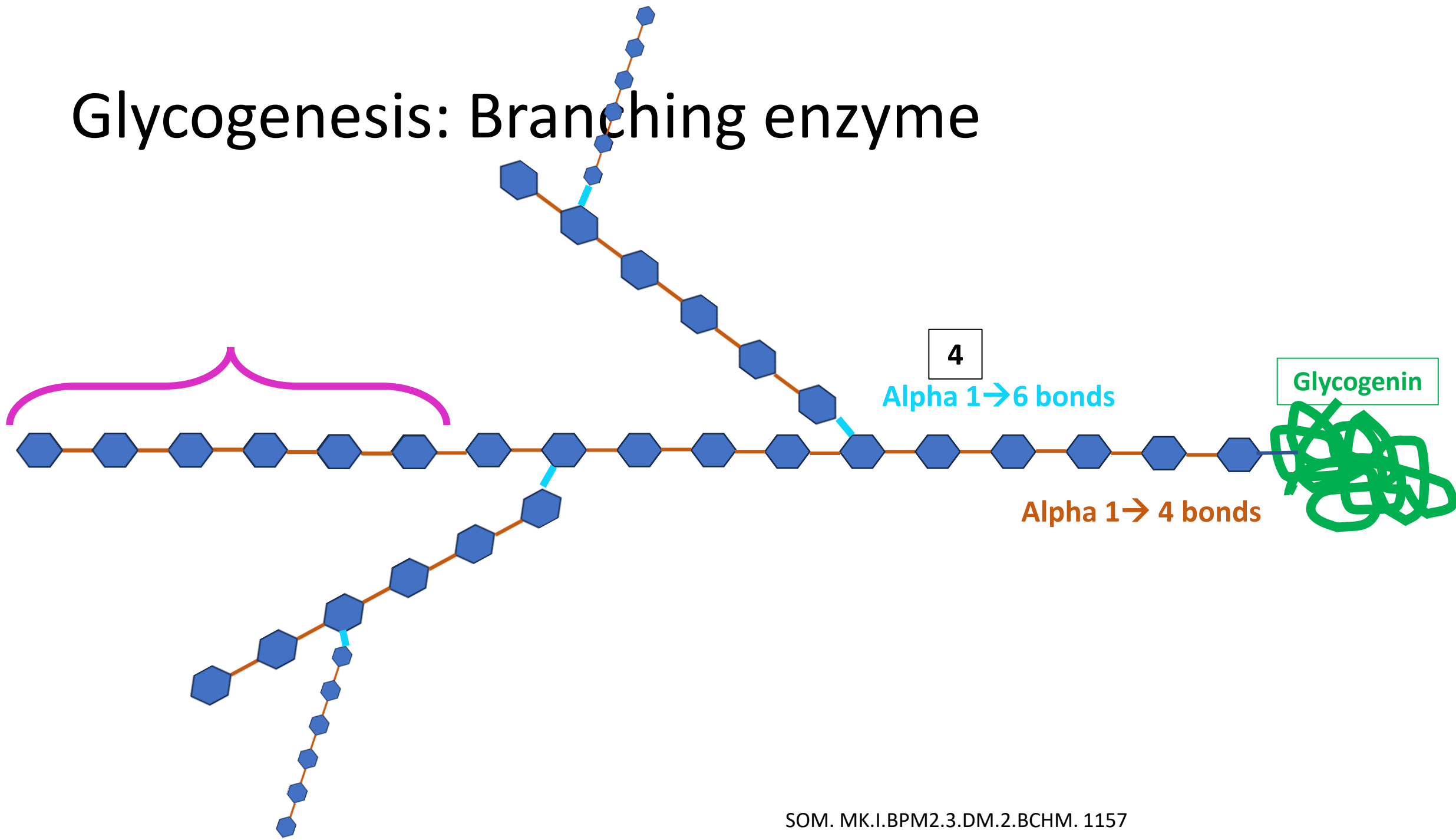


Glycogenesis: Glycogen synthase

- **Regulated enzyme**
 - Active in fed state
 - **High insulin: glucagon ratio** → **Dephosphorylation (Active form)**
- Repeatedly adds glucose units to form long glucose polymers (chains)
- Forms **alpha 1→4** glycosidic links
- Glycogenin (primer protein) if there is no pre-existing glycogen molecule



Glycogenesis: Branching enzyme

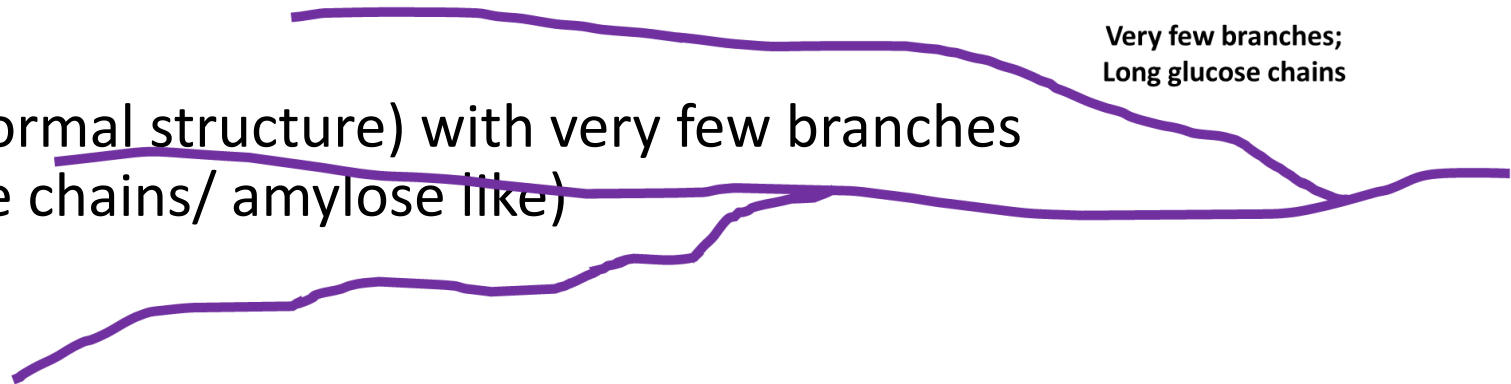


Branching enzyme

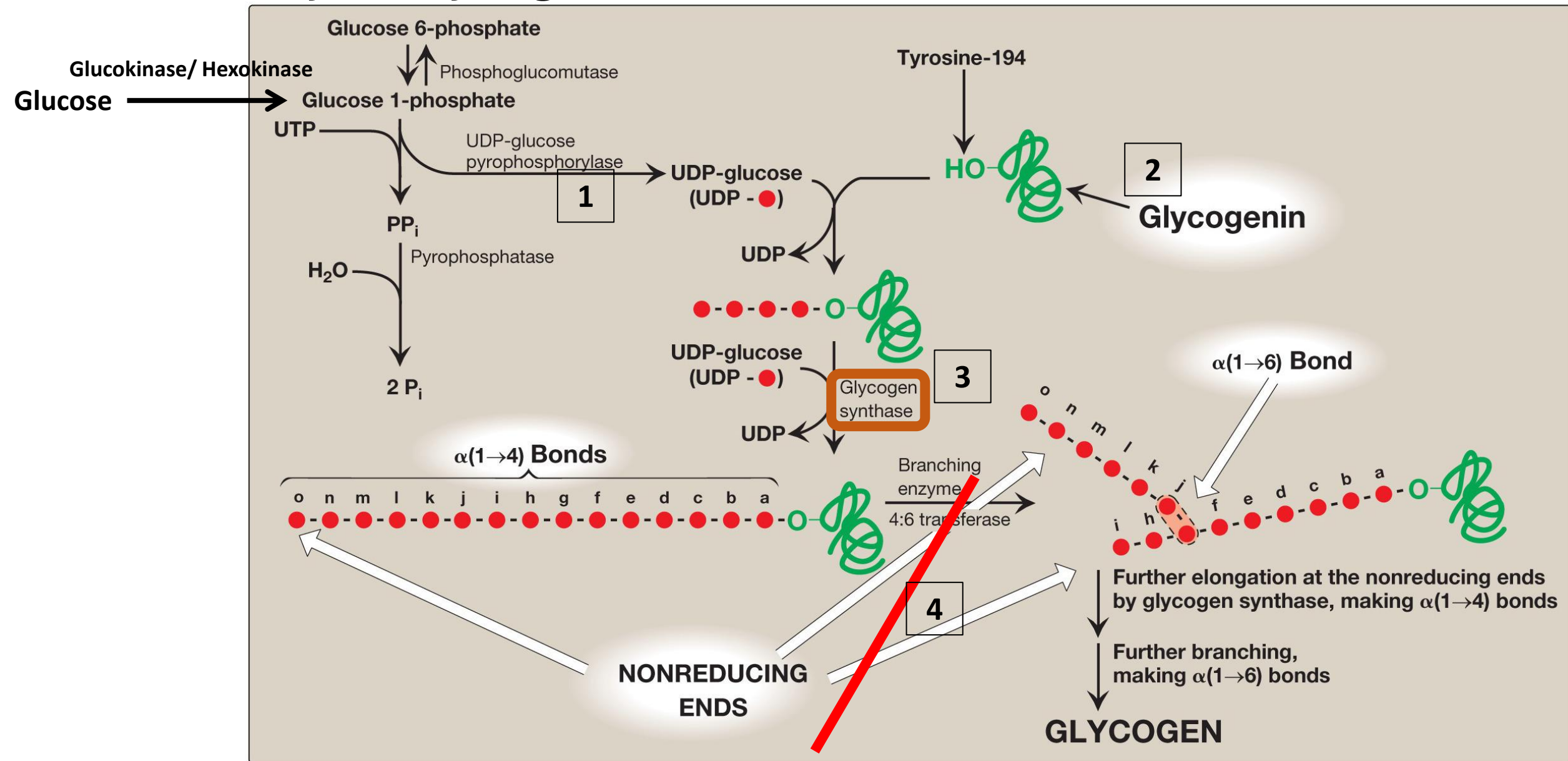
- Creates **alpha 1→6 bonds** (branch points)
- Repeated action of glycogen synthase and branching enzyme → Highly branched glycogen structure
- Transfers 6-8 glucose units (Cuts 1→4 bonds) and creates **1→6 bonds** (Aka 4:6 transferase)

- **Andersen disease:**

- Branching enzyme deficiency
- Accumulation of Glycogen (abnormal structure) with very few branches (glycogen with very long glucose chains/ amylose like)
- Liver damage and cirrhosis



Summary of Glycogenesis

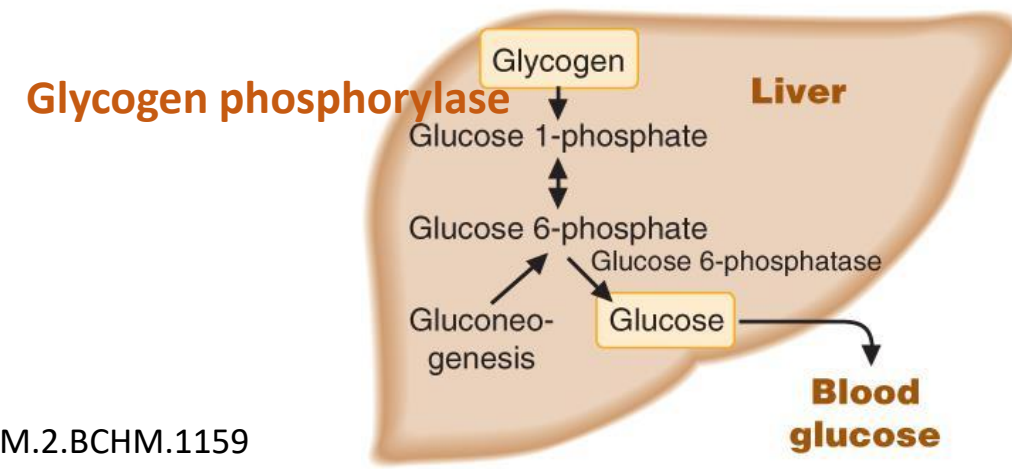
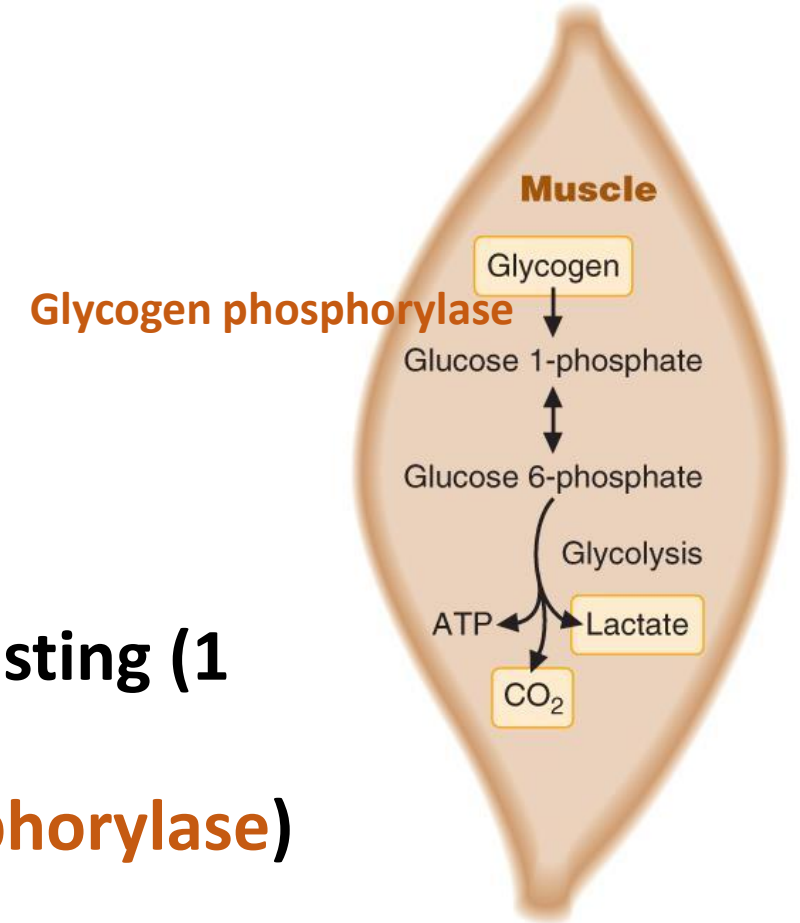


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Andersen disease

Glycogenolysis

- [Metabolism | Glycogenolysis – YouTube](#)
- **Different in muscle and liver** (different functions)
 - Different fates of glucose 6-phosphate
 - Regulation is different
- **Liver glycogen stores depleted by 17-24 hours of fasting (1 day)**
- Removal of glucose as Glucose 1-P (**Glycogen phosphorylase**)
 - Regulated enzyme
 - Conversion to Glucose 6-P
- Removal of branches (**Debranching enzyme**)
- Fate of glucose 6-P
 - Liver (**Glucose 6-phosphatase**)
 - Muscle

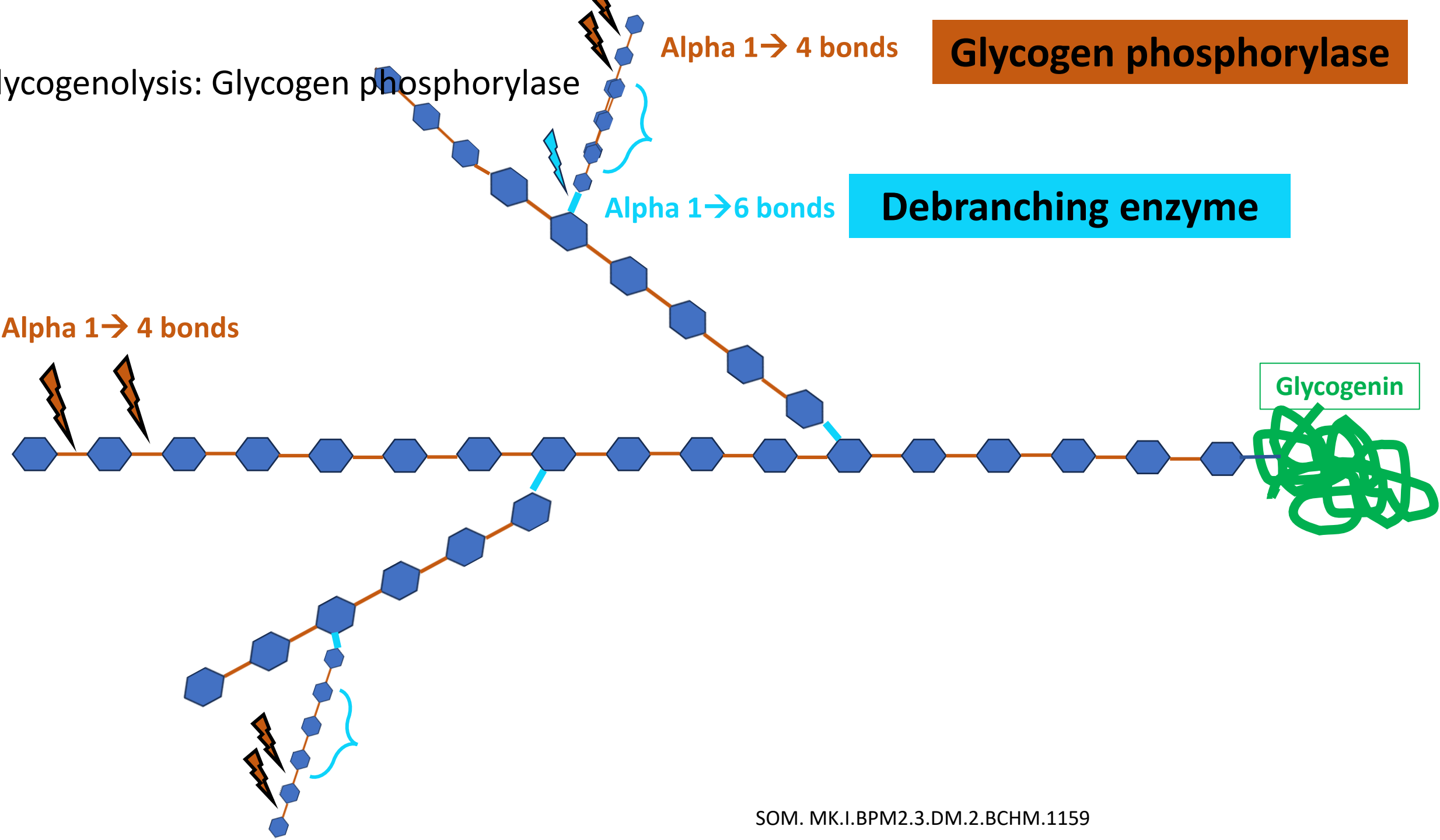


Glycogenolysis: Glycogen phosphorylase

Glycogen phosphorylase

Debranching enzyme

Glycogenin

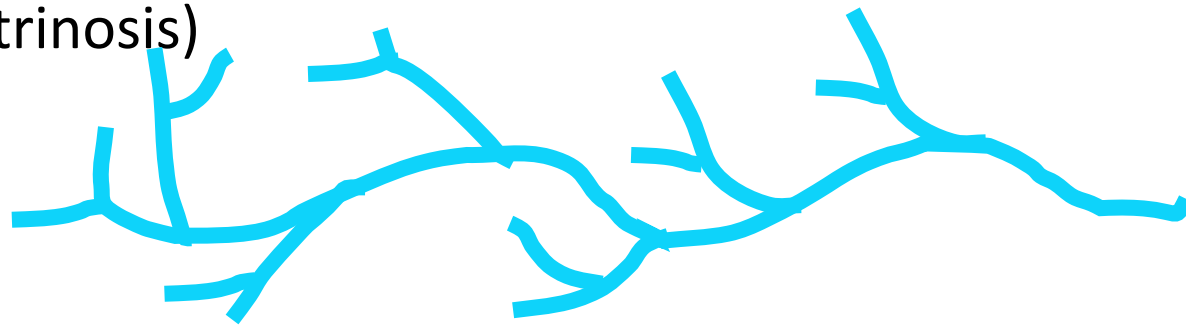


Glycogenolysis: **Glycogen phosphorylase**

- Acts on ends of branches to release glucose as glucose 1-P
- Cuts alpha 1→4 links
- Glucose 6-P formed by phosphoglucomutase
- Glycogen phosphorylase acts until the branch is about 4 glucose units long
- Regulated enzyme
- **McArdle disease: (Type V GSD)**
 - Muscle glycogen phosphorylase deficiency
 - Inability to perform anaerobic exercise (cramps during exercise)
 - Blood lactate levels do NOT increase following exercise
- **Hers disease: (Type VI GSD)**
 - Liver glycogen phosphorylase deficiency
 - Hypoglycemia and increased liver glycogen stores

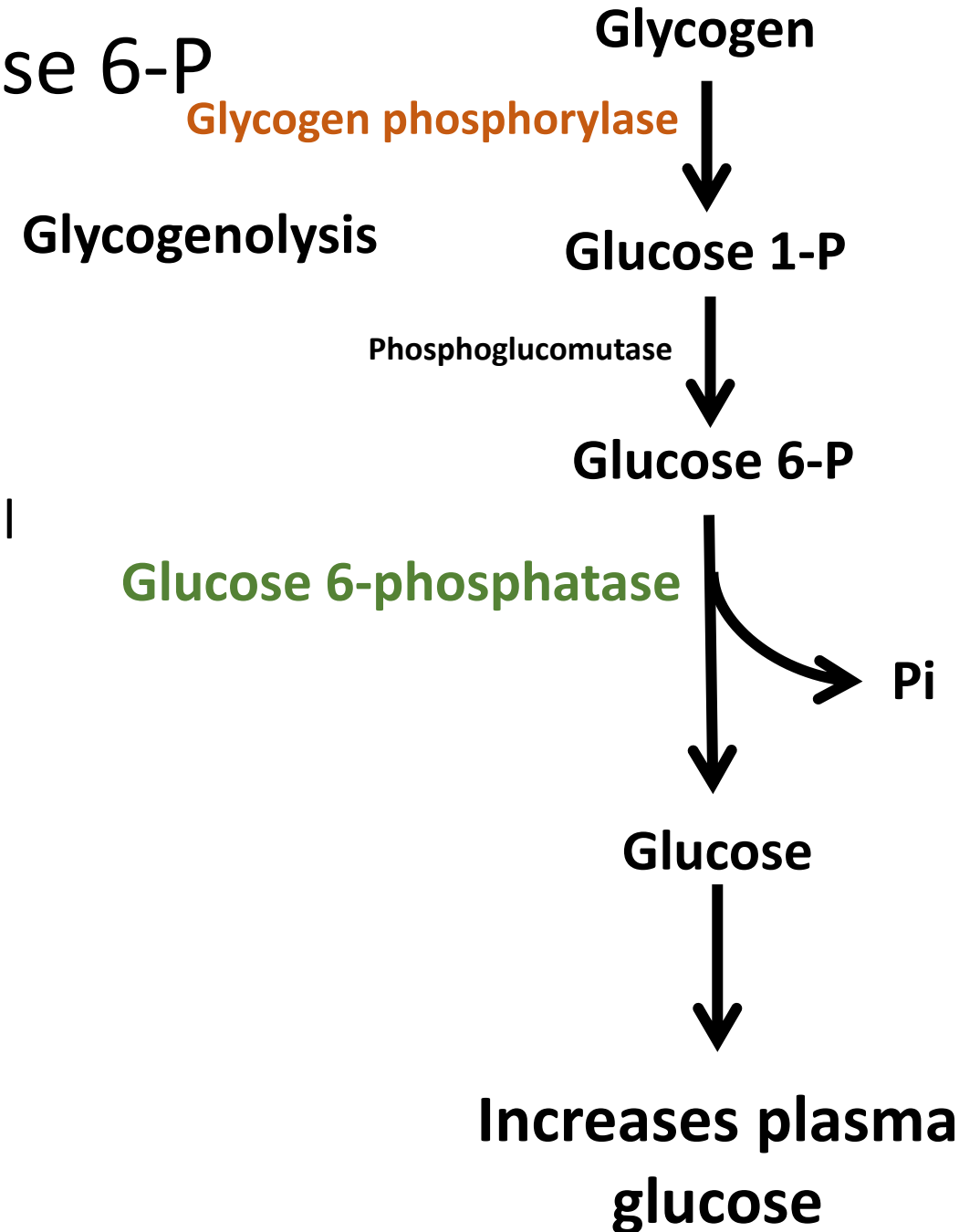
Glycogenolysis: Debranching enzyme

- Removal of branches
 - Cuts alpha 1→4 links (moves 3 glucose units to neighboring branch) and makes new alpha 1→4 link
- **Cleaves alpha 1→6 link** to release free glucose
- Repeated action of glycogen phosphorylase and debranching enzyme
- Cori disease: (Type III GSD)
 - Debranching enzyme deficiency
 - Hypoglycemia and muscle disease
 - Accumulation of glycogen (abnormal structure) with short outer branches (limit dextrin) (Limit dextrinosis)



LIVER Glycogenolysis: Fate of glucose 6-P

- **Glucose 6-phosphatase**
 - Endoplasmic reticulum: **Liver and renal tubular cells**
 - Also involved in **gluconeogenesis**
- Forms free glucose → Increases plasma glucose level
- **Von Gierke disease (type I GSD):** Most common
 - Glucose 6-phosphatase deficiency
 - Severe hypoglycemia (Defect in glycogenolysis and gluconeogenesis)
 - Hepatomegaly: Excess Glycogen accumulation (structure normal)
 - Muscle NOT affected



Reciprocal regulation of glycogen metabolism: **LIVER**

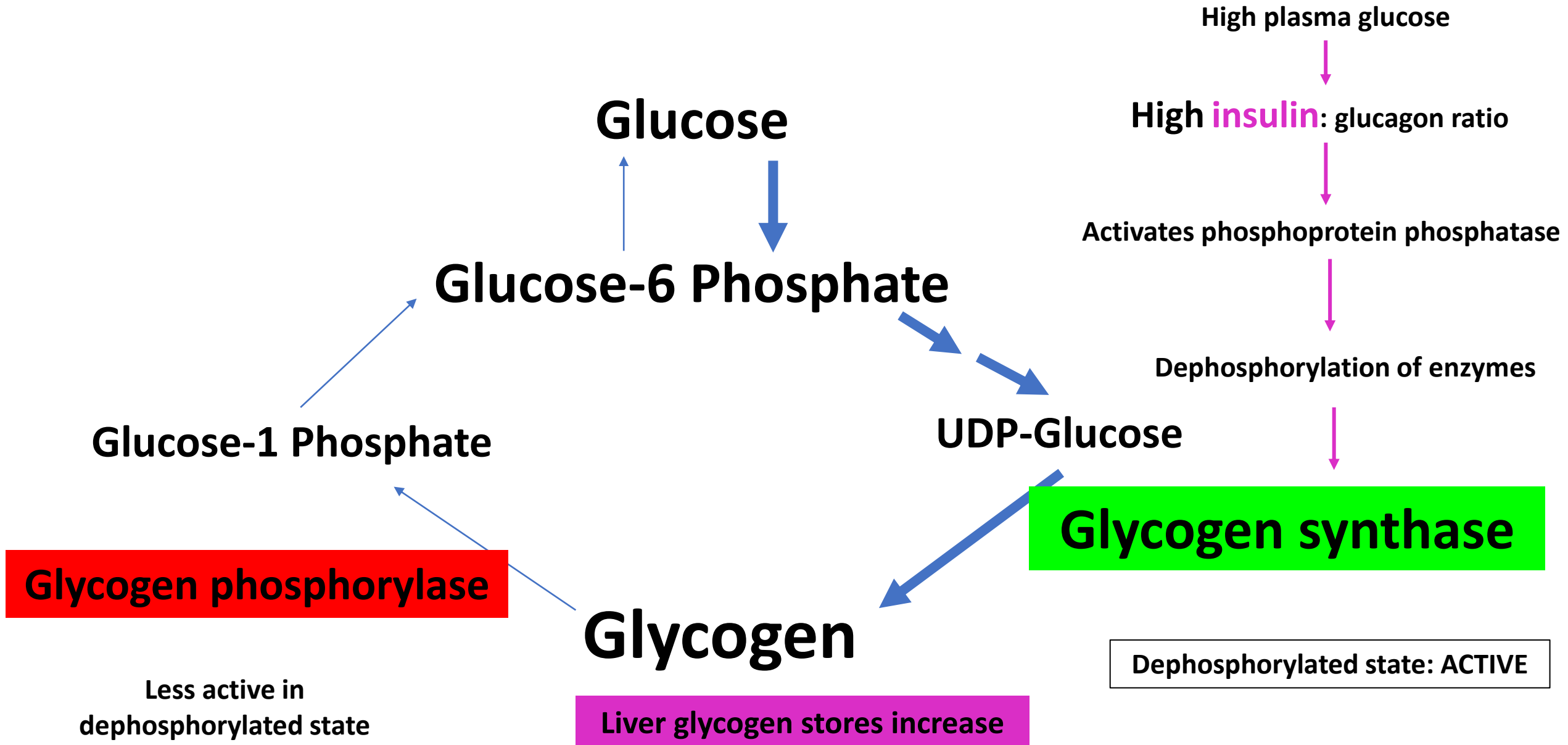
Glycogen synthesis (Glycogenesis)

- Regulated enzyme: Glycogen synthase
- Active in fed state
 - High insulin: glucagon ratio
 - Active phosphoprotein phosphatase
 - Dephosphorylation of enzyme

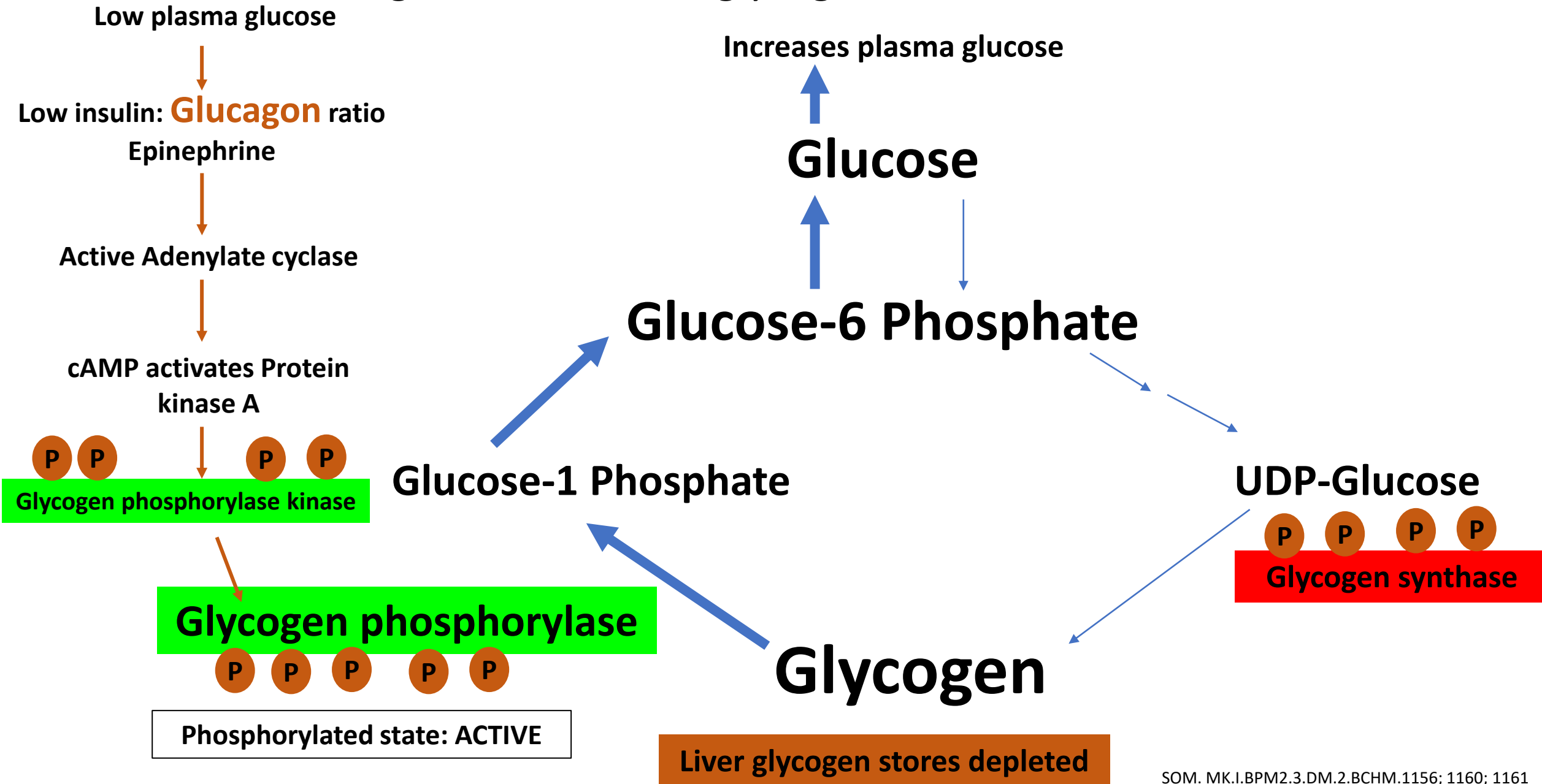
Glycogen degradation (Glycogenolysis)

- Regulated enzyme: Glycogen phosphorylase
- Active in fasting state
 - Low insulin: glucagon ratio
 - Epinephrine
 - Adenylate cyclase $\rightarrow \uparrow \text{cAMP} \rightarrow$ Active Protein kinase A
 - Phosphorylation of enzyme
 - Glycogen phosphorylase kinase and glycogen phosphorylase

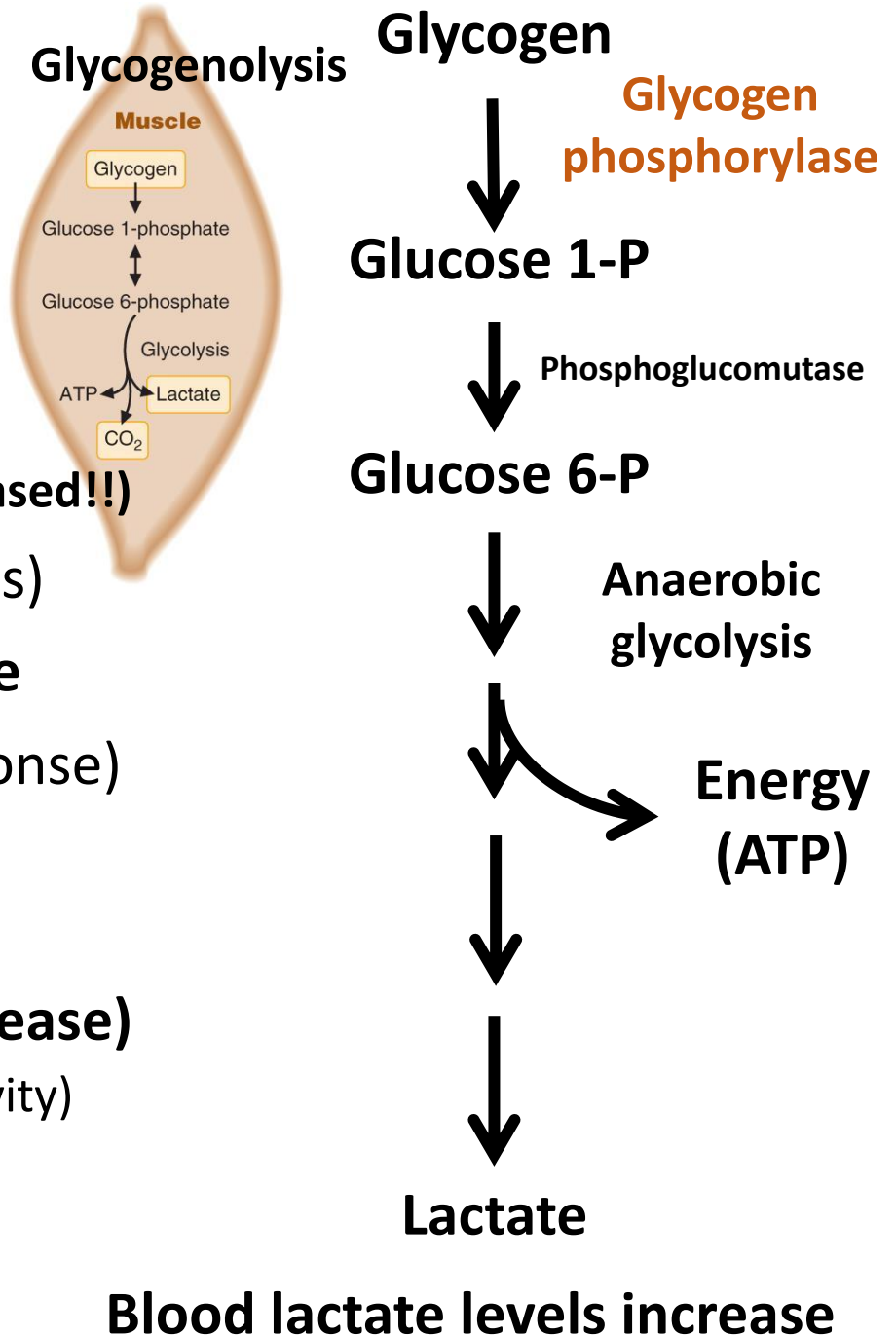
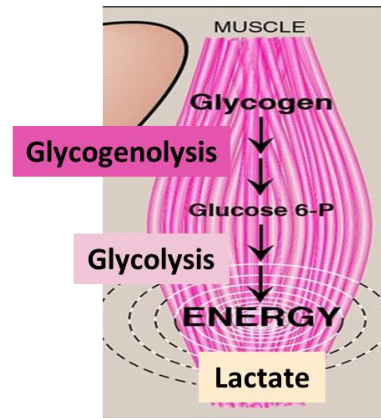
Regulation of **LIVER** glycogen metabolism: FED state



Regulation of **LIVER** glycogen metabolism: FASTING state



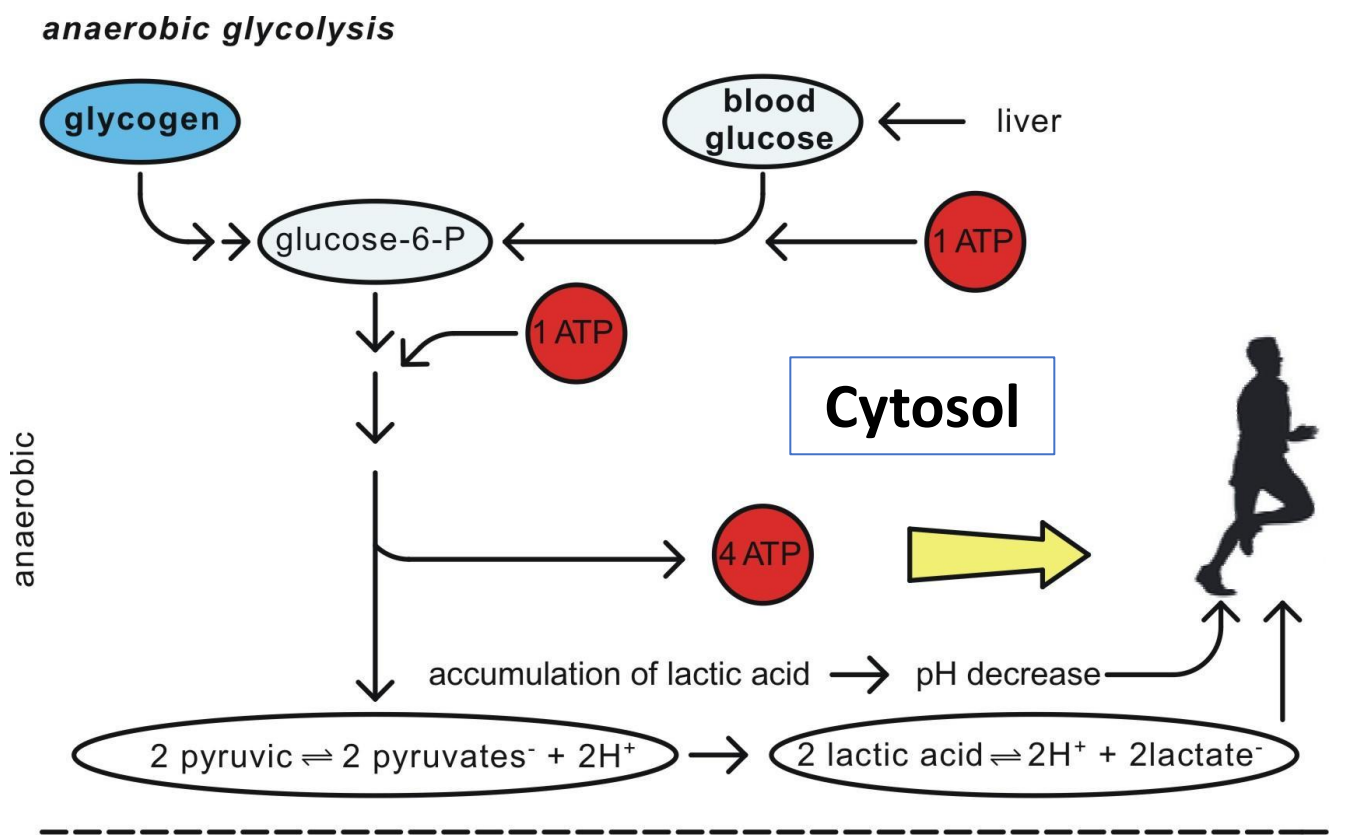
MUSCLE Glycogenolysis: Fate of glucose 6-P



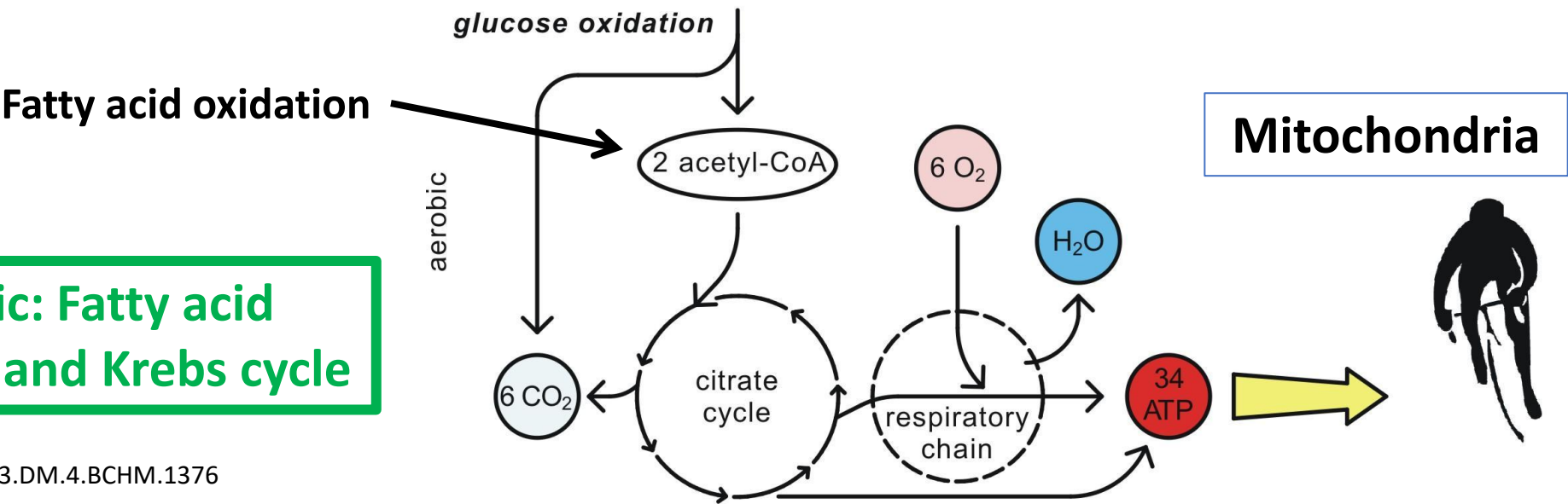
- Glucose 6-phosphatase NOT present in muscle (free glucose not released!!)
- Glycogen → Glucose 6-P → Lactate (via anaerobic glycolysis)
- **Glycogenolysis and glycolysis forms lactate during exercise**
- During exercise, blood lactate levels increase (normal response)
- Forearm exercise test:
 - Blood lactate levels increase (normal response)
- Muscle glycogen storage disorders: **(McArdle and Tarui disease)**
 - Cramps on exercise (unable to perform anaerobic/ strenuous activity)
 - Cramps due to low Energy (Low ATP)
 - **NO increase in blood lactate following forearm exercise test**
 - Tarui disease (PFK-1 deficiency)

Metabolism
in muscle:
Aerobic and
anaerobic

Glycogenolysis and
Glycolysis (Anaerobic)



Aerobic: Fatty acid
oxidation and Krebs cycle



Regulation of glycogen metabolism: **MUSCLE**

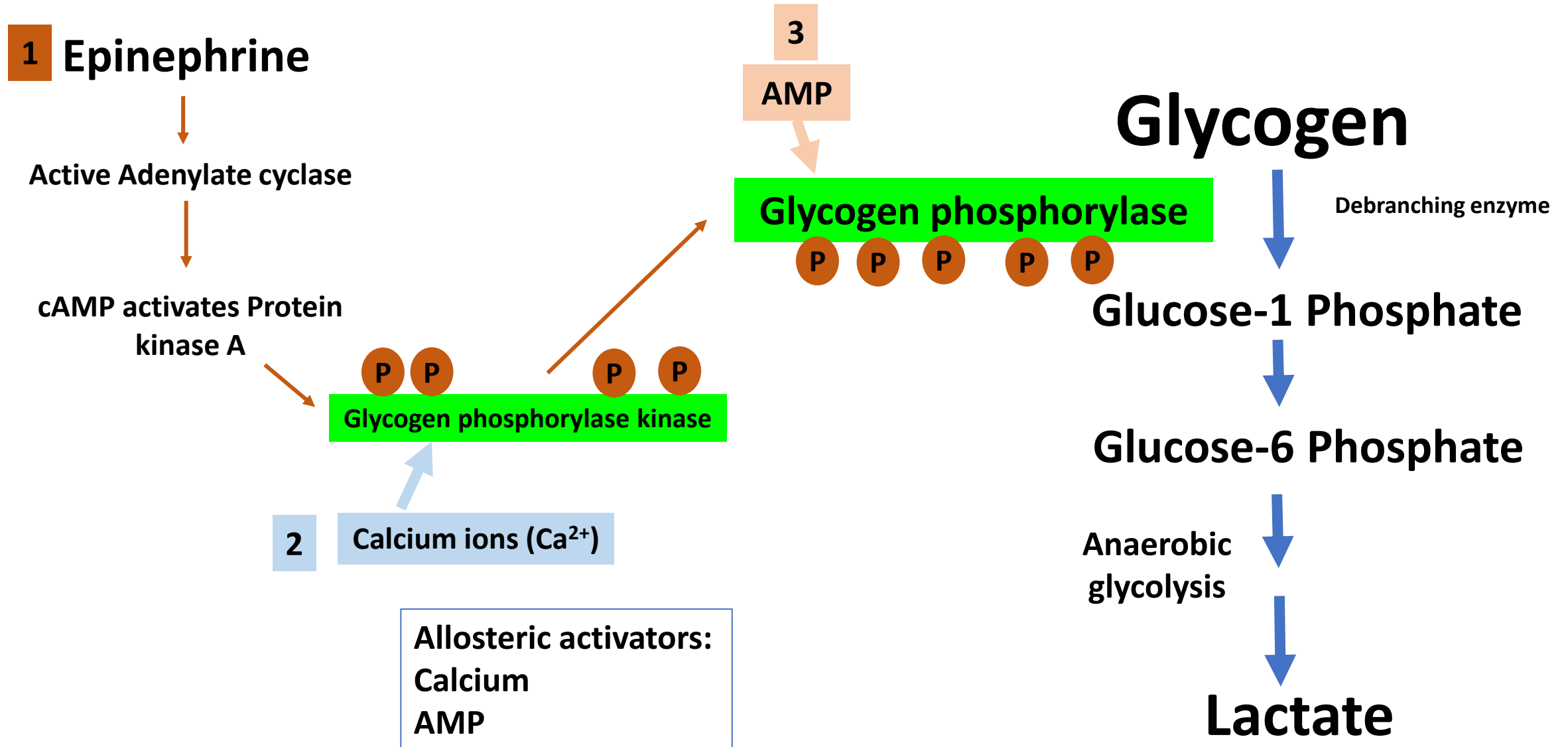
Glycogen synthesis (Glycogenesis)

- Regulated enzyme: Glycogen synthase
- Active in fed state
 - High insulin: glucagon ratio
 - Active phosphoprotein phosphatase
 - Dephosphorylation of enzyme

Glycogen degradation (Glycogenolysis)

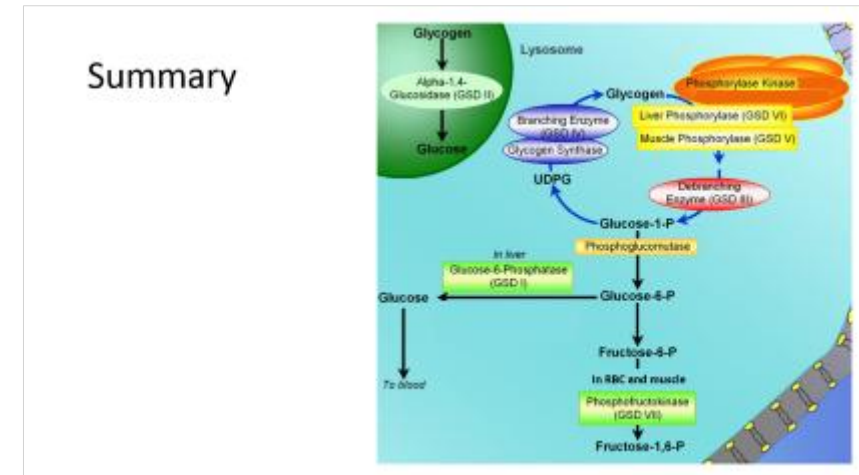
- Regulated enzyme: Glycogen phosphorylase
- Active during **exercise/ stress**
 1. **Epinephrine** activates
 - Adenylate cyclase $\rightarrow \uparrow$ cAMP $\rightarrow$ Active Protein kinase A
 - Phosphorylation of enzyme
 - Glycogen phosphorylase kinase and glycogen phosphorylase
 2. **Calcium** activates glycogen phosphorylase kinase
 3. **AMP** activates Glycogen phosphorylase

Regulation of **MUSCLE** glycogenolysis: EXERCISE



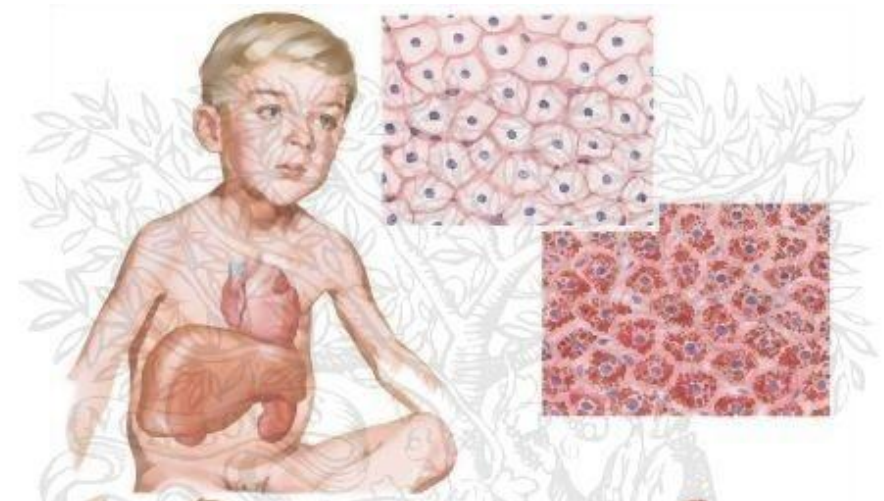
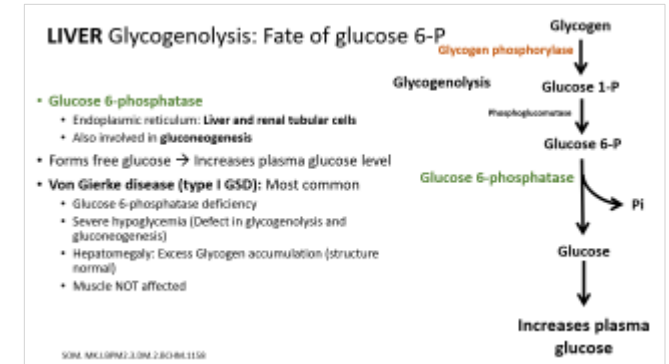
Glycogen storage disorders (GSD)

- Inherited autosomal recessive disorders
- Excessive glycogen accumulation
- **Liver glycogen storage disorders**
 - Von Gierke disease (Type I)
 - Cori disease (Type III) – Muscle also affected
 - Andersen disease (Type IV)
 - Hers disease (Type VI)
- **Muscle glycogen storage disorders**
 - McArdle disease (Type V)
 - Tarui disease (Type VII)
- **Lysosomal glycogen storage disorder: Pompe disease (Type II)**



Von Gierke disease (Type I GSD)

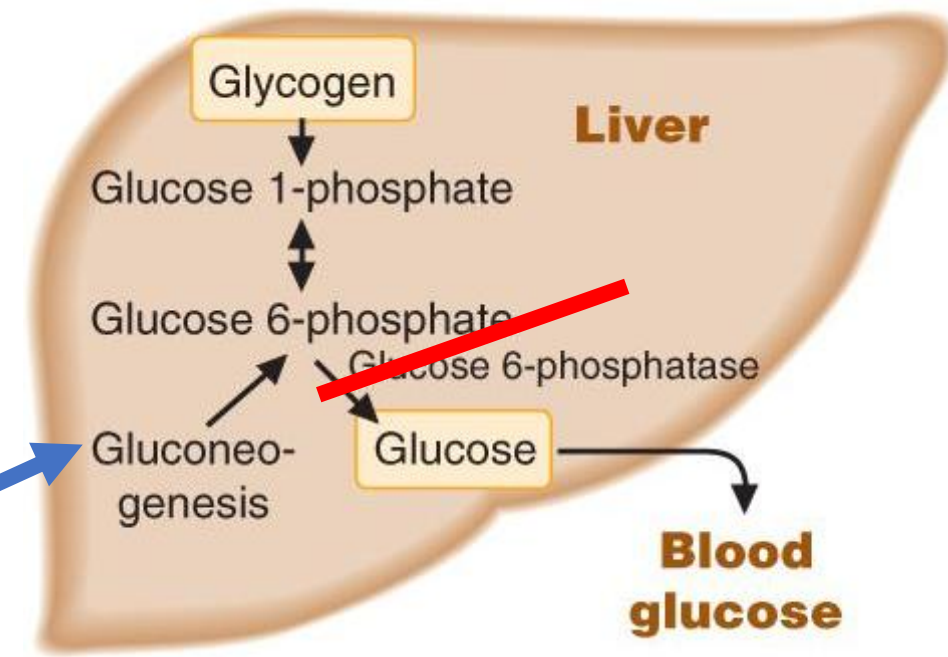
- Most common; 6 months to 2 years
- Glucose 6-Phosphatase deficiency (Liver and kidney)
- **Hypoglycemia:** Impaired Liver glycogenolysis and gluconeogenesis → Severe fasting hypoglycemia (35-50mg/dL; 6-8 hours after meal) → Drowsy, Lethargy, seizures
- **Hepatomegaly:** Accumulation of glucose 6-P → Stimulates glycogen synthesis and excessive glycogen stores in LIVER and KIDNEY
- **Lactic acidosis:**
 - Metabolic acidosis: Low pH; Low Bicarbonate; Low PCO₂
 - Lactate from peripheral tissues is NOT converted to glucose in liver (CORI cycle impaired)
 - Excessive glucose 6-P in liver forms lactate



Von Gierke disease (Type I GSD)

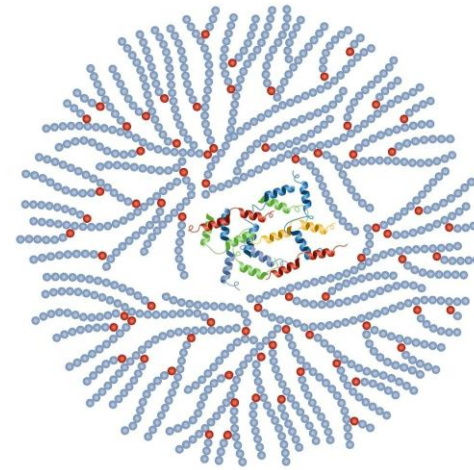
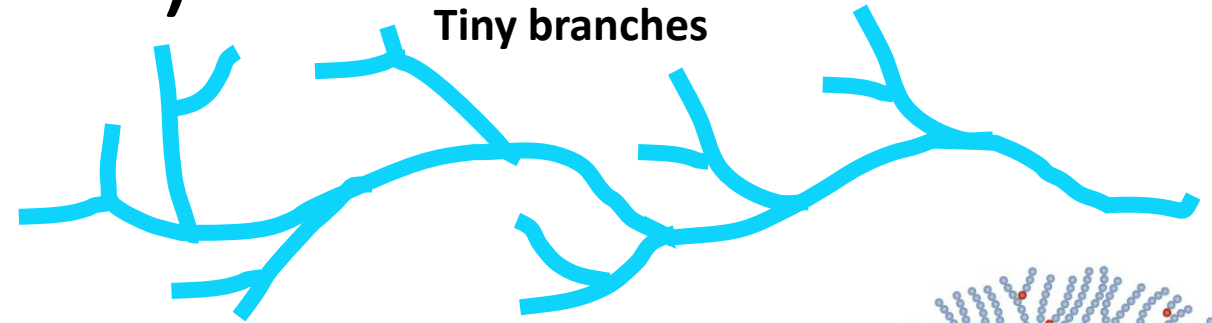
- Growth failure and 'doll-like face'
- Uric acid levels elevated
 - Competes with lactate in renal tubule
 - Increased Glucose 6-P → Ribose 5-P → Purine nucleotide turnover → Increased Uric acid
- Elevated serum triglyceride levels
- Injection of glucagon → NO increase in plasma glucose
- Management:
 - Prevent hypoglycemia: Avoid fasting; Continuous infusion of uncooked starch
 - [First Patient Receives Gene Therapy for Rare Liver Disorder - YouTube](#)

Lactate

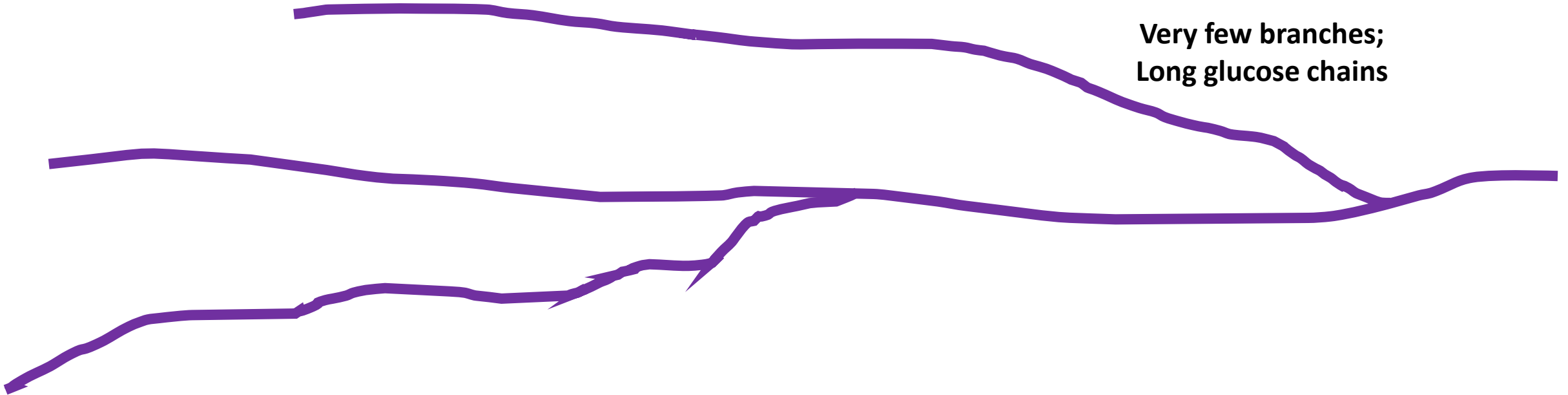


Cori disease (Type III GSD)

- Debranching enzyme deficiency (Glycogenolysis defect)
- Hypoglycemia and muscle disease (many patients)
 - Hypoglycemia not as severe as in type I as gluconeogenesis is not affected
- **Liver and muscle biopsy:** Accumulation of glycogen (abnormal structure) with **short outer branches (limit dextrin)** (Limit dextrinosis)



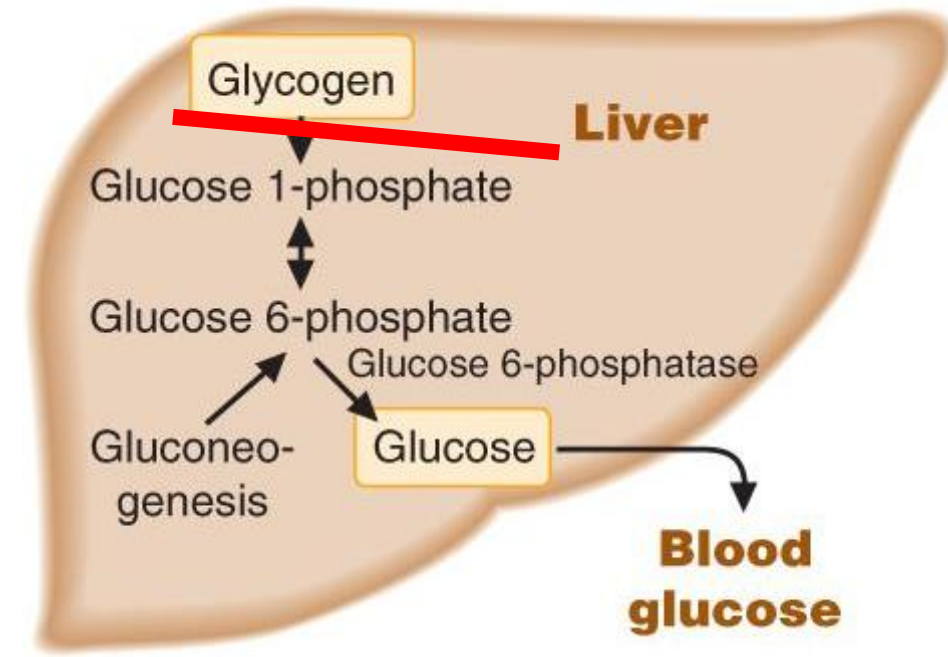
Andersen disease (Type IV GSD)



- Branching enzyme deficiency (Defect in Glycogen synthesis)
- **Liver and muscle biopsy:** Accumulation of Glycogen (abnormal structure) with **very few branches** (glycogen with **very long glucose chains/ amylose like**)
- **Progressive liver damage and cirrhosis** (Abnormal liver function tests) → Fatal in childhood

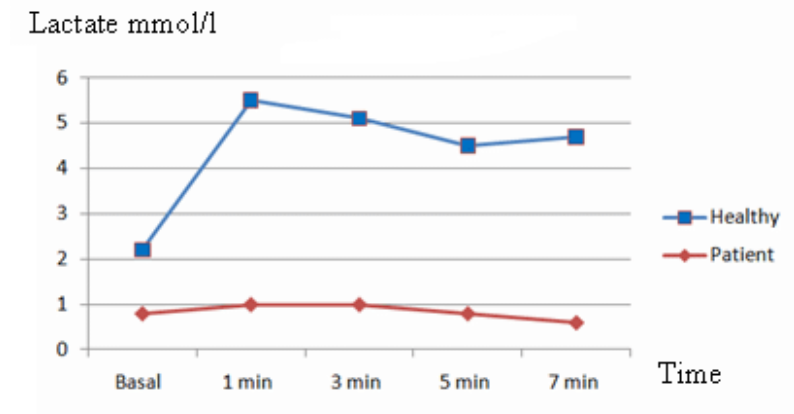
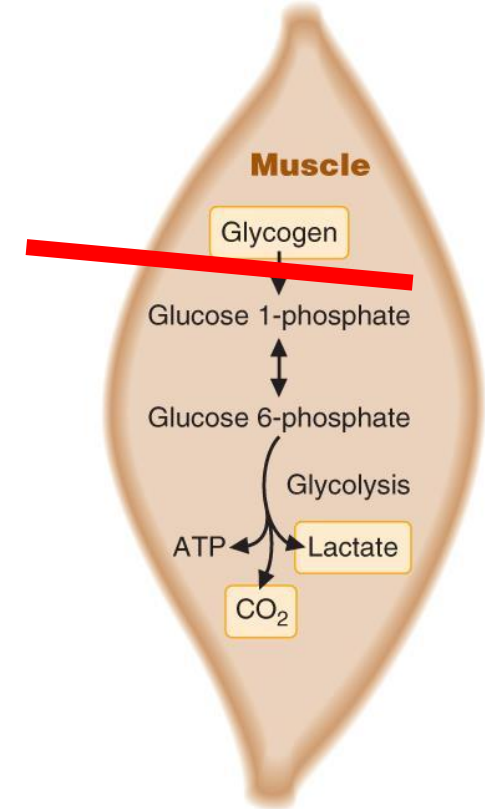
Hers disease (Type VI GSD)

- Deficiency of liver glycogen phosphorylase
- Mild hypoglycemia
 - Gluconeogenesis is not impaired
- Hepatomegaly due to glycogen accumulation



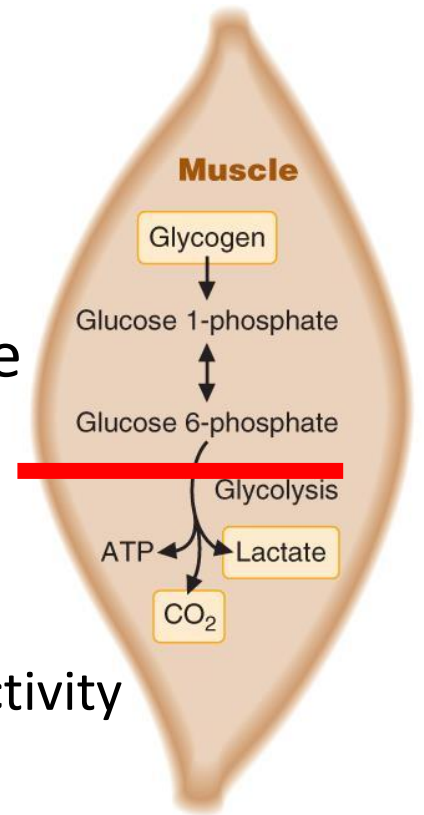
McArdle disease (Type V GSD)

- Poor exercise tolerance, cramps, and muscle pain during or after exercise
- Rhabdomyolysis → Myoglobin in urine (Brown urine)
- **Muscle glycogen phosphorylase deficiency**
- **“Second wind phenomenon”**: Improved exercise tolerance when low-grade activity continued
 - Aerobic fatty acid oxidation due to oxygenation
- Lab tests:
 - Elevated CK (CK-MM): non-specific marker for muscle damage
 - Myoglobinuria (rhabdomyolysis)
 - Forearm exercise test: **NO increase in lactate following strenuous exercise**
 - Muscle biopsy: Accumulation of glycogen with normal branching structure
 - NO hypoglycemia/hepatomegaly/cardiomegaly/ hemolysis



Tarui disease (Type VII GSD)

- Deficiency of muscle **Phosphofructokinase-1 (PFK-1)** (Defect in muscle glycolysis)
- Presentation similar to McArdle:
 - Poor exercise tolerance, cramps and muscle pain during or after exercise
 - Rhabdomyolysis → Myoglobin in urine (Brown urine)
 - **“Second wind phenomenon”**: Improved exercise tolerance when low-grade activity continued (Aerobic fatty acid oxidation due to oxygenation)
- Lab tests:
 - Elevated CK (CK-MM): non-specific marker for muscle damage
 - Myoglobinuria: non-specific marker for muscle damage
 - Forearm exercise test: **NO increase in lactate following strenuous exercise**
 - Muscle biopsy: Accumulation of glycogen with normal branching structure
 - NO hypoglycemia/hepatomegaly/ cardiomegaly
 - **Mild hemolysis**



Differentiate between McArdle and Tarui

Pompe disease (Type II GSD)

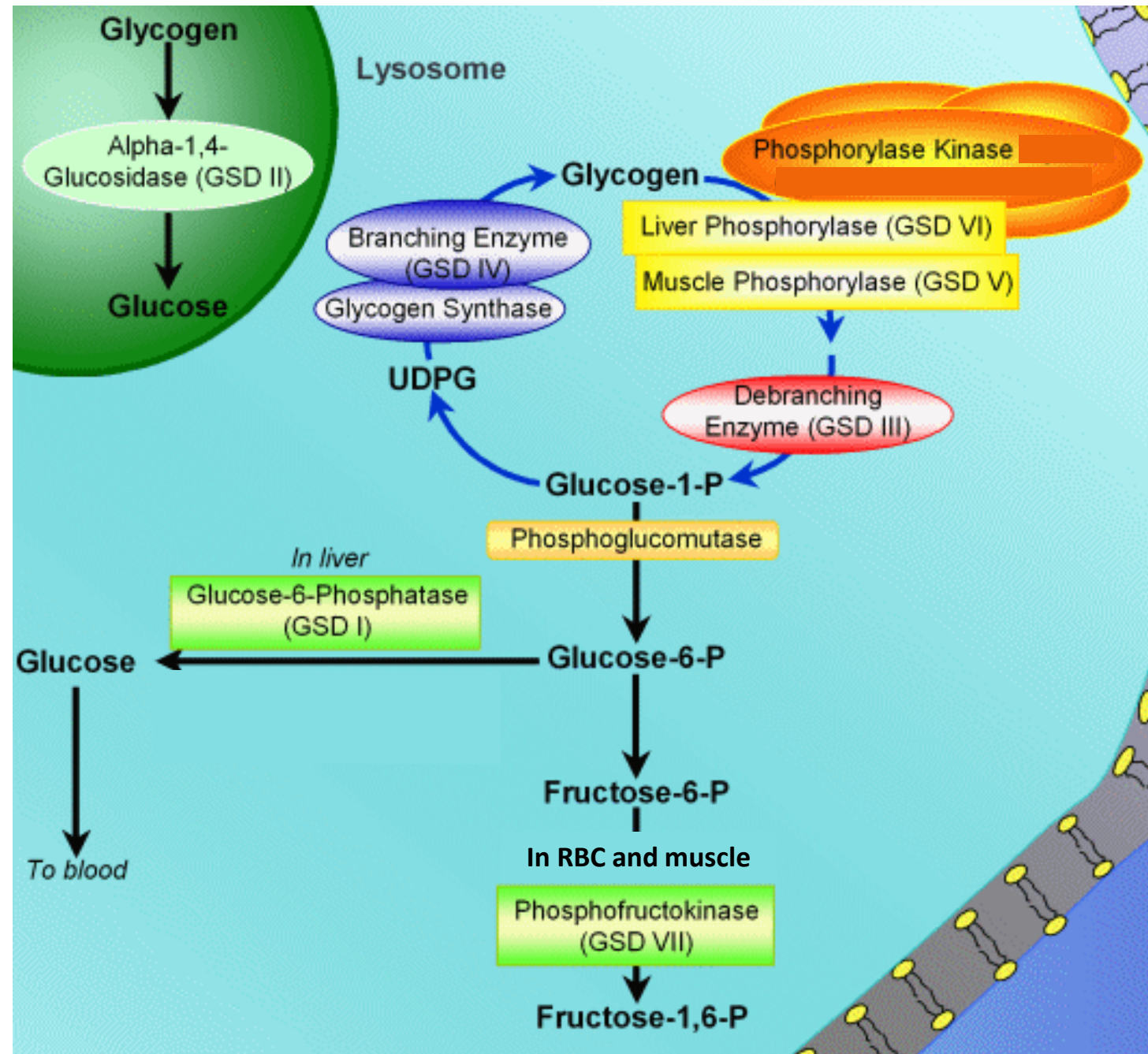
- Lysosomal α -glucosidase (Acid maltase)
- <https://rarediseases.org/rare-diseases/pompe-disease/#rdb-video>
- Pompe disease (Type II GSD)
 - Lysosomal glycogen accumulation
 - **Cardiac muscle affected: Cardiomyopathy**
 - Skeletal muscle weakness and hypotonia
 - Respiratory failure
 - Macroglossia
 - NO HYPOGLYCEMIA
 - **Management: Enzyme replacement therapy**

Floppy infant
(Hypotonia)



Example of a 15-month-old child with T2PD showing marked facial hypertrophy and macroglossia and dilated

Summary



Summary

Hepatic

Myopathic

Generalized

Disease (Presentation age)	Enzyme deficiency	Lab findings	Clinical features
GSD I: von Gierke disease (3 months – 2 years)	Glucose 6-phosphatase (ER)	Severe fasting HYPOGLYCEMIA; LACTIC ACIDOSIS (Metabolic acidosis with high anion gap); hyper triglyceridemia; hyperuricemia Liver biopsy: Excess glycogen of normal structure (normal branching pattern)	Growth failure (doll face); twitching and seizures; Hepatonephromegaly (glycogen deposits)
GSD II: Pompe disease (6 months to older)	Acid maltase (Lysosomal glucosidase)	NO HYPOGLYCEMIA ; Lysosomal glycogen deposits in heart, skeletal muscle and liver	Cardiomegaly ; Floppy infant (low muscle tone)
GSD III: Cori disease (toddlers to older children)	Debranching enzyme	Hypoglycemia Liver (and muscle) biopsy: Excess glycogen with short branches	Hepatomegaly (Limit dextrin/abnormal glycogen) Progressive muscle weakness
GSD IV: Andersen disease (1-2 years)	Branching enzyme (4:6 transglycosylase)	Elevated ALT and AST (Liver damage) Liver (and muscle) biopsy: Glycogen with very long branches (very few branches)	Hepatomegaly → Cirrhosis (5-7 years) Hypoglycemia
GSD V: McArdle disease (Adolescents)	Muscle glycogen phosphorylase (Myophosphorylase)	Forearm ischemic exercise test: NO increase in Lactate levels (Anaerobic exercise); Elevated CK levels and myoglobin in urine (Muscle damage) NO hypoglycemia Muscle biopsy: Excess glycogen of normal structure (normal branching pattern)	Muscle cramps and fatigue during exercise Rhabdomyolysis and brown urine
GSD VI: Hers disease (Children)	Liver glycogen phosphorylase	Hypoglycemia and hepatomegaly Liver biopsy: Excess glycogen of normal structure (normal branching pattern)	Hypoglycemia and hepatomegaly
GSD VII: Tarui disease (Adolescents)	Muscle Phosphofructokinase-1	Forearm ischemic exercise test: NO increase in Lactate levels (Anaerobic exercise); Elevated CK levels and myoglobin in urine (Muscle damage); Hemolysis (elevated LDH levels); NO hypoglycemia Muscle biopsy: Excess glycogen of normal structure (normal branching pattern)	Muscle cramps and fatigue during exercise Rhabdomyolysis and brown urine